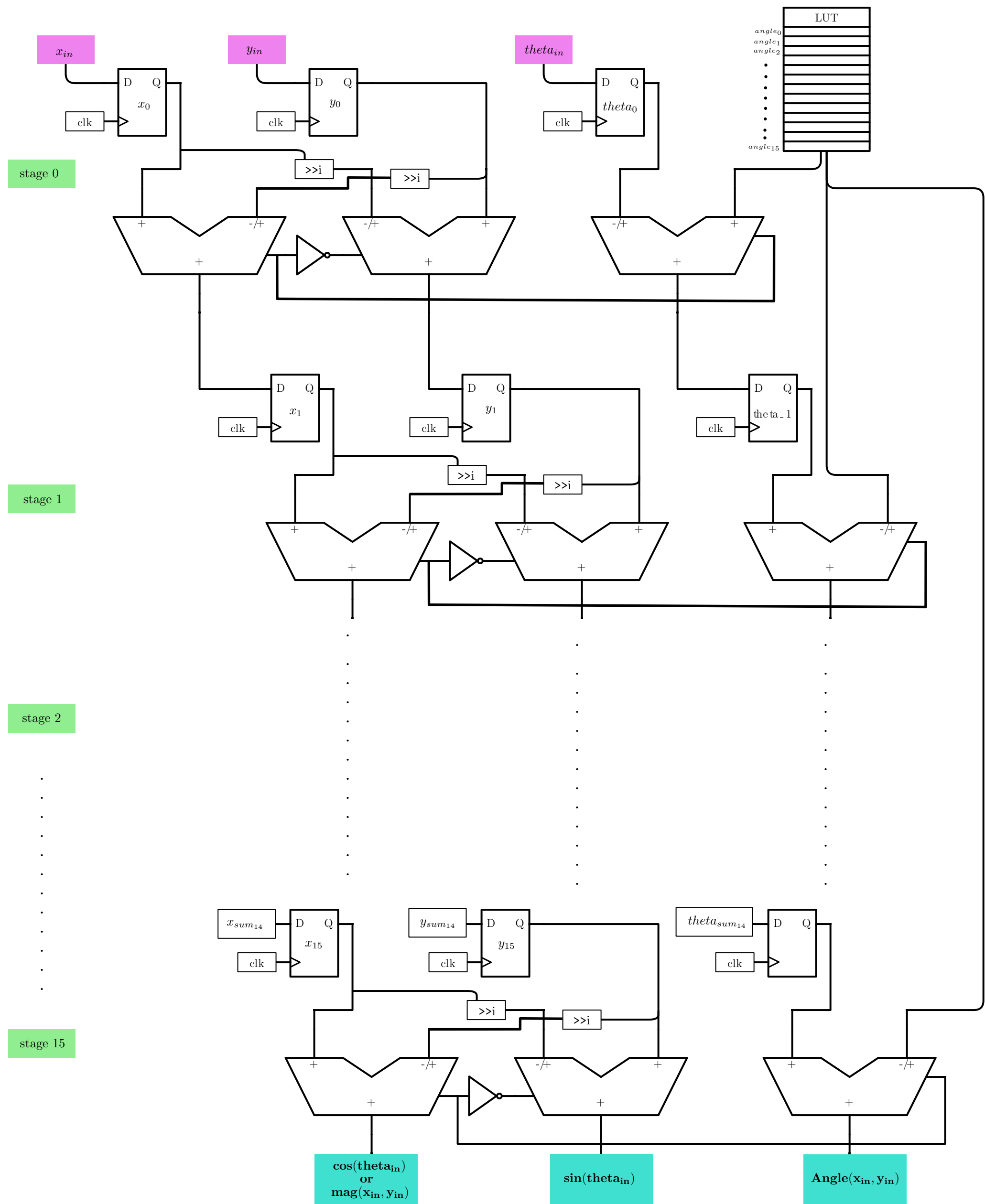


CORDIC 16-Stage Pipe-lined Architecture

mode = 0 : cosine, sine (Rotation Mode)
mode = 1 : Magnitude, Angle (Vectoring Mode)
>>i : Barrel Shifter



mode = 0

Theta MSB = 0 >>

$$\begin{aligned} x_{k+1} &= x_k - 2^{-k} * y_k \\ y_{k+1} &= y_k + 2^{-k} * x_k \\ theta_{k+1} &= theta_k - angle \end{aligned}$$

Theta MSB = 1 >>

$$\begin{aligned} x_{k+1} &= x_k + 2^{-k} * y_k \\ y_{k+1} &= y_k - 2^{-k} * x_k \\ theta_{k+1} &= theta_k + angle \end{aligned}$$

mode = 1

y MSB = 0 >>

$$\begin{aligned} x_{k+1} &= x_k + 2^{-k} * y_k \\ y_{k+1} &= y_k - 2^{-k} * x_k \\ theta_{k+1} &= theta_k + angle \end{aligned}$$

y MSB = 1 >>

$$\begin{aligned} x_{k+1} &= x_k - 2^{-k} * y_k \\ y_{k+1} &= y_k + 2^{-k} * x_k \\ theta_{k+1} &= theta_k - angle \end{aligned}$$

Latency: 16 Clock Cycles
Throughput: 1 Clock Cycle